COMPUTER ENGINEERING
QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 212 \| Automating Server Management | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 9/24/26 | **DATE SUBMITTED:** | 9/24/26 |
| **NAME:** | Villareal, Vanderohe Vernon | **INSTRUCTOR:** | Engr. Jimlord Quejado |

## MIDTERM SKILLS EXAM

### 1. PROCEDURE OUTPUT

Task 1: Prepare the Server

**Creating a directory for the skills exam on the control node.**

```
villareal1@workstation:~$ mkdir Midterm_Exam
villareal1@workstation:~$ cd Midterm_Exam
```

**Verify that ansible is on the managed nodes**

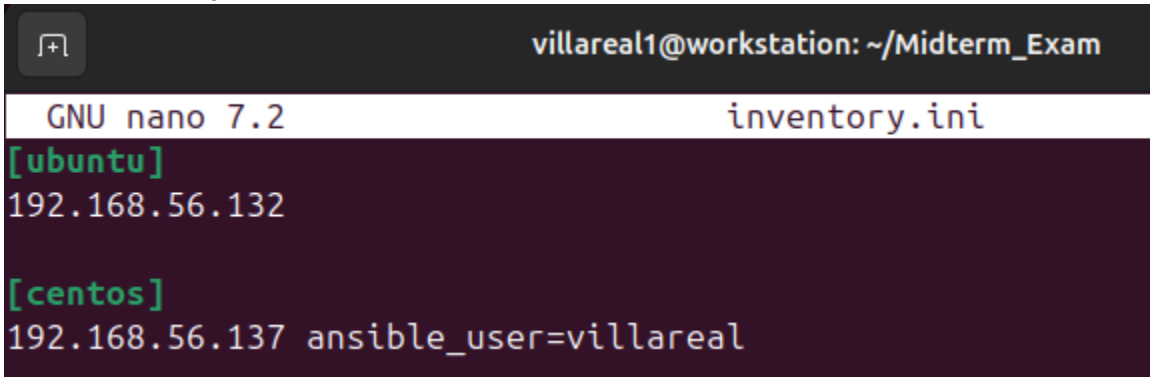
CentOS

```
[villareal@localhost ~]$ ansible --version
ansible [core 2.14.18]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/villareal/.ansible/plugins/modules', '
/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /home/villareal/.ansible/collections:/usr/share/
ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Aug 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Re
d Hat 11.5.0-14)] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

Ubuntu

```
villareal1@server1:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/villareal1/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/villareal1/.ansible/collections:/usr/share
/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

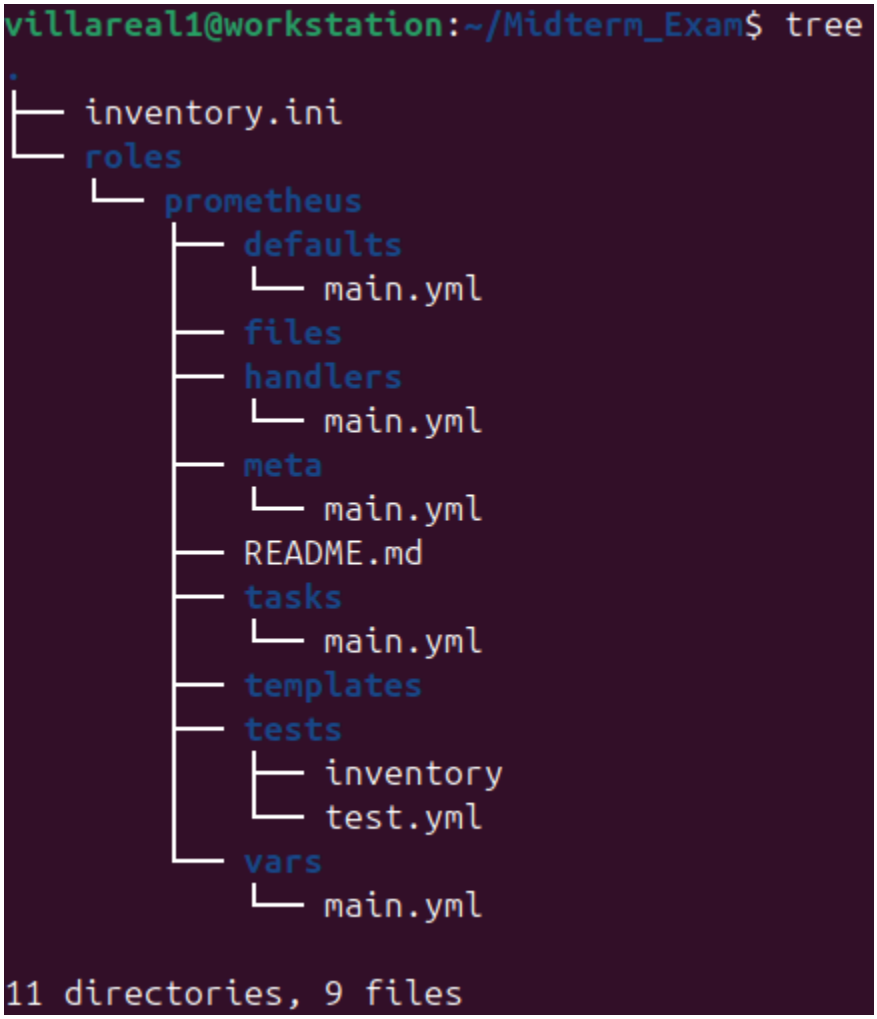
**Create an inventory on the control node.**




**Verify connections**

```
villareal1@workstation:~/Midterm_Exam$ ansible all -m ping -i inventory.ini
192.168.56.132 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.56.137 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
```

**Create roles on the control node.**


villareal1@workstation:~/Midterm_Exam$ tree
.
├── inventory.ini
└── roles
    └── prometheus
        ├── defaults
        │   └── main.yml
        ├── files
        ├── handlers
        │   └── main.yml
        ├── meta
        │   └── main.yml
        ├── README.md
        ├── tasks
        │   └── main.yml
        ├── templates
        ├── tests
        │   ├── inventory
        │   └── test.yml
        └── vars
            └── main.yml
11 directories, 9 files

```
  GNU nano 7.2                    roles/prometheus/tasks/main.yml
---
- name: installing dependencies for ubuntu
  ansible.builtin.apt:
    name:
      - wget
      - curl
      - tar
      - gzip
    state: present
  when: ansible_os_family == "Debian"

- name: installing dependencies for centos
  ansible.builtin.dnf:
    name:
      - wget
      - curl
      - tar
      - gzip
    state: present
  when: ansible_os_family == "RedHat"
```

```
TASK [prometheus : installing dependencies for centos] ************************
ok: [192.168.56.137]
```

**Task 2: Create the Prometheus User**

```
- name: Create prometheus groups
  ansible.builtin.group:
    name: prometheus
    state: present

- name: Create prometheus user
  ansible.builtin.user:
    name: prometheus
    group: prometheus
    state: present
```

Ubuntu

```
TASK [prometheus : Create prometheus groups] ***********************************
ok: [192.168.56.132]

TASK [prometheus : Create prometheus user] *************************************
ok: [192.168.56.132]
```

Centos

```
TASK [prometheus : Create prometheus groups] ***********************************
ok: [192.168.56.137]

TASK [prometheus : Create prometheus user] *************************************
ok: [192.168.56.137]
```

**Task 3: Create Prometheus Directories**

```
- name: Create prometheus directory
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
```

Ubuntu

```
TASK [prometheus : Create prometheus directory] ********************************
ok: [192.168.56.132]
```

Centos

```
TASK [prometheus : Create prometheus directory] ********************************
ok: [192.168.56.137]
```

**Task 4: Download Prometheus and Extract**

```
- name: Download prometheus for managed nodes
  ansible.builtin.get_url:
    url: https://github.com/prometheus/prometheus/releases/download/v2.55.1/pro
    dest: /tmp/prometheus-2.55.1.linux-amd64.tar.gz
    mode: '0755'

- name: Extract prometheus
  ansible.builtin.unarchive:
    src: /tmp/prometheus-2.55.1.linux-amd64.tar.gz
    dest: /tmp
    remote_src: yes
```

Ubuntu

```
TASK [prometheus : Download prometheus for managed nodes] **********************
ok: [192.168.56.132]

TASK [prometheus : Extract prometheus] *****************************************
changed: [192.168.56.132]
```

Centos

```
TASK [prometheus : Download prometheus for managed nodes] **********************
ok: [192.168.56.137]

TASK [prometheus : Extract prometheus] *****************************************
ok: [192.168.56.137]
```

**Task 5: Install the Prometheus Binaries**

```
- name: Install prometheus binaries
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64.tar.gz
    dest: /usr/local/bin/prometheus
    remote_src: yes
    owner: root
    group: root
    mode: '0755'
```

Ubuntu

```
TASK [prometheus : Install prometheus binaries] ********************************
changed: [192.168.56.132]
```

Centos

```
TASK [prometheus : Install prometheus binaries] ********************************
ok: [192.168.56.137]
```

**Task 6: Install the Prometheus Configuration Files**

Ubuntu

Centos

**Task 7: Validate the Configuration**

Ubuntu

Centos

**Task 8: Create the systemd Service**

Ubuntu

Centos

**Task 9: Start and Enable Prometheus**

Ubuntu

Centos

**Task 10: Verify Prometheus**
Ubuntu

Centos

**Task 11: Access Prometheus Through a Browser**
Ubuntu

Centos

## 2. CONCLUSION

**In conclusion, this midterm skills exam made us think critically as we redo most hands-on activities from scratch. This skills examination improved my way of thinking which would most likely help in the future as a student and as a professional. This also helped me in being more competent in resolving errors on my own, since there are difficulties that I have encountered back when I was doing the past hands-on activities. In short , I learned a lot on this skills exam where it would most likely help me in the future, not only as a responsible student but also as a better thinker.**

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **NONE**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **NONE**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **NONE**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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